Python – Weekly Assignment Answers

Simple explanations and beginner-friendly programs

1. Name the main built-in data types used in Python.

The main built-in Python data types are:

Numeric: int, float, complex

Text: str

Sequence: list, tuple, range

Mapping: dict

Set: set, frozenset

Boolean: bool (True or False)

Binary: bytes, bytearray, memoryview

None type: NoneType, represented by None

2. Explain the difference between integer, float and complex data types.

int: Whole numbers without a decimal point, such as 7 or -4.

float: Numbers with a decimal point, such as 3.5 or -0.25.

complex: Numbers with real and imaginary parts, such as 2+3j.

```
a = 7       # int
b = 3.5     # float
c = 2 + 3j  # complex
```

3. What is type conversion? Explain implicit and explicit conversion.

Type conversion means changing a value from one data type to another.

Implicit conversion: Python converts a value automatically when appropriate.

```
x = 5
y = 2.5
result = x + y  # result is 7.5 (float)
```

Explicit conversion: The programmer converts the type using functions such as int(), float() or str().

```
x = "12"
y = int(x)  # y is the integer 12
```

4. Write a program to accept two numbers and display their sum, difference, product and quotient.

```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))

print("Sum:", a + b)
print("Difference:", a - b)
print("Product:", a * b)

if b != 0:
    print("Quotient:", a / b)
else:
    print("Quotient is undefined because division by zero is not allowed.")
```

5. Write a program to convert temperature from Celsius to Fahrenheit.

Formula: Fahrenheit = (Celsius × 9/5) + 32

```
celsius = float(input("Enter temperature in Celsius: "))
fahrenheit = (celsius * 9 / 5) + 32
print("Temperature in Fahrenheit:", fahrenheit)
```

6. Write a program to calculate the area and perimeter of a rectangle.

Area = length × breadth; Perimeter = 2 × (length + breadth).

```
length = float(input("Enter length: "))
breadth = float(input("Enter breadth: "))

area = length * breadth
perimeter = 2 * (length + breadth)
```

```
print("Area:", area)
print("Perimeter:", perimeter)
```

7. Write a program to calculate simple interest using principal, rate and time.

Formula: Simple Interest = (Principal × Rate × Time) / 100

```
principal = float(input("Enter principal amount: "))
rate = float(input("Enter annual rate (%): "))
time = float(input("Enter time in years: "))

simple_interest = (principal * rate * time) / 100
print("Simple Interest:", simple_interest)
```

8. Explain arithmetic, comparison, logical and assignment operators.

Arithmetic operators: Perform calculations: +, -, *, /, //, %, **.

Comparison operators: Compare values and return True or False: ==, !=, >, <, >=, <=.

Logical operators: Combine or reverse conditions: and, or, not.

Assignment operators: Assign or update a variable: =, +=, -=, *=, /=, //=, %=, **=.

Example: x = 5; x += 2 updates x to 7.

9. What is the difference between division and floor division?

Division (/): Gives the ordinary division result, usually including a decimal part.

Floor division (//): Rounds the quotient down to the nearest whole-number value.

```
print(7 / 2)   # 3.5
print(7 // 2)  # 3
print(-7 // 2) # -4
```

10. What is operator precedence? Why is it important?

Operator precedence is the order in which Python evaluates operators in an expression.

It is important because it determines the result when an expression contains different operators. Parentheses can make the intended order clear.

```
result = 2 + 3 * 4
print(result)  # 14, multiplication first

result = (2 + 3) * 4
print(result)  # 20, parentheses first
```